Ankita Mishra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.72
Rani Benode Manjari Govt. Girls' School	Class XII (West Bengal Council of Higher Secondary Education)	2018	83.80%
Rani Benode Manjari Govt. Girls' School	Class X (West Bengal Board of Secondary Education)	2016	89.86%

PROJECTS

Low-Latency Limit Order Book & Matching Engine [GitHub]

- Built a **header-only C++ limit order book and matching engine** with **price-time priority**, supporting **limit, market, IOC, fill-or-kill and post-only** orders, plus cancel and modify with **exchange-style queue-priority rules**.
- Validated correctness with unit tests, **invariant checks after every operation**, and a reproducible **randomized fuzzer (5M+ operations)** that verifies the emitted **event stream exactly reconstructs the book's state**.
- Built a **benchmark harness** on a realistic **1M-operation, 10,000-order workload** and **profiled the 68.8 ns/op baseline**, attributing **~49% of runtime to heap allocation**, ~17% to std::map traversal and ~15% to hash-table operations; used this to drive a **pre-allocated order pool** with an **intrusive free list** and **handle-linked FIFO queues**.

FABRIC: Federated Adaptive BiomedCLIP with Instance-level Clinical Prompts [GitHub]

- Implemented the **FACMIC** paper - a **federated learning framework,** which uses a **feature attention module** and **domain adaptation** to classify medical images while keeping client data private.
- **Novelty :** Integrated a **LLaMA 3.2** pipeline to dynamically generate visual-radiological prompts, eliminating static intra-class text limitations. Along with this we replaced the standard **CLIP** backbone with **BioMedCLIP** (trained on the medical domain), achieving an accuracy improvement of over **~4%** compared to the baseline implementation.

Comparative ML Classifier Study [GitHub]

- Evaluated 7 classifiers **(Logistic Regression, SVM, Naive Bayes, Decision Tree, Random Forest, AdaBoost, XGBoost)** on two real-world datasets: Breast Cancer Wisconsin **(balanced)** and Telco Customer Churn **(imbalanced, mixed-type data)**.
- Performed **EDA**, **multicollinearity/VIF analysis**, **SHAP interpretability**, and PCA-based decision boundary visualization.
- Compared **class-imbalance handling techniques** including class-weighting, sample-weighting,scale-pos-weight, and threshold tuning across classifiers, evaluating their impact on recall,precision, F1-score, and overall performance across both datasets.
- Selected the best-fit model for each dataset using business-relevant metrics, prioritizing recall while balancing precision,F1-score, and overall performance for reliable classification on imbalanced datasets.

Diabetic Retinopathy Severity Grading using Deep Learning [GitHub]

- Built a 5-class DR severity grading pipeline on the **APTOS 2019** dataset using a fine-tuned **EfficientNetB3** CNN with class-balanced sampling to handle severe data imbalance, evaluated using **Quadratic Weighted Kappa (QWK)** as the metric, achieving a Test QWK of **0.82**.
- Applied **Grad-CAM** to visually verify the model's predictions were driven by genuine retinal features, and implemented **Monte Carlo Dropout**-based uncertainty quantification for selective prediction, raising QWK to **~0.90** on the most confident 80% of cases.
- Deployed the trained model as an interactive **Streamlit** application with real-time Grad-CAM visualization and uncertainty-based specialist-referral flagging.

Company Financial Annual Report Analysis Agent [GitHub]

- Built an **agentic RAG system** using **LangGraph and Mistral LLM**, with a multi-tool agent that extracts structured

financial data from annual reports into a hybrid **SQL + vector database**, enabling multi-company trend/CAGR analysis and grounded natural-language Q&A.
- Engineered a **self-verifying agent pipeline** with automated **fact-grounding**, **self-critique**, and validated by offline tests that detected real hallucination.
- **Deployed** the system as a **FastAPI** backend + **Streamlit** frontend, **containerized with Docker**, with an automated **CI pipeline** (**GitHub Actions**) for testing and automated Docker image builds.

WORK EXPERIENCE

Wipro Limited - Kolkata, *Project Engineer*

Upskilled in C++ development through rigorous technical training.

SKILLS

- **Programming Languages:** C, C++, Python, SQL
- **Systems & Linux:** Linux, Processes, Threads, Process Scheduling, System Calls, Inter-Process Communication, Synchronization
- **CPU Microarchitecture:** Cache Hierarchy, Memory Hierarchy, Instruction Pipelining, Branch Prediction
- **Networking:** TCP/IP, TCP Congestion Control, HTTP/1.1, HTTP/2, QUIC, NAT, DHCP, Wireshark
- **Algorithms & Complexity:** Data Structures, Algorithms, Time & Space Complexity
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, Transformers, Federated Learning
- **GenAI/LLM:** LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering
- **Tools & Technologies:** GitHub, Jupyter Notebook, Google Colab, LaTeX/Overleaf

RELEVANT COURSES

Foundation of Machine Learning | Deep Learning | Visual Computing | Representation Learning | Advanced Computer Networks | Advance Data Structure and Algorithm

SCHOLASTIC ACHIEVEMENTS

Awarded the **Silver Medal** for securing **2nd Rank** in **Bachelor of Technology (B.Tech)** in **Computer Science & Engineering** at **JIS University** for academic excellence.

PUBLICATIONS

Facemask Detection and Maintaining Safe Distance Using AI and ML to Prevent COVID-19--A Study, ICRCCV-2023 [link]

POSITIONS OF RESPONSIBILITY

Represented the **Department of Computer Science and Engineering, JIS University, Kolkata**, during my **Bachelor of Technology (B.Tech.)** at **TechX19**, an inter-college technical exhibition organized by **Adamas University, Kolkata**.